

1) Why the two answers differ

The two statements correspond to two different optimization targets:

Objective A (what the automated report used): “Robust deployability for short-premium”

This objective prioritizes:

- **Robustness / diversification of evidence**
- **Avoiding single-factor fragility**
- **High agreement with peers** (but not necessarily perfect SHAP agreement)
- **Interpretability** as “good enough,” not maximal

That objective picks **epoch 3** (because it has the best balance: lowest top-feature dominance + highest entropy, while still having strong surrogate fidelity and strong top-K agreement).

Objective B (what I accidentally switched to): “Interpretability consistency across methods”

This objective prioritizes:

- **SHAP vs permutation agreement at the top** as a hard requirement, because it signals explanation-method invariance.

That objective tends to pick **epoch 4** (because it’s the only one with perfect Top-5 overlap between SHAP and permutation).

These objectives are both rational — but they are not the same.

2) Lock the objective function (so “best” is definitional, not ambiguous)

Let each model $M \in \{E3, E4, E5\}$ have these measured properties:

- Surrogate fidelity: $R^2(M)$
- Attribution entropy: $H(M)$
- Top-feature dominance: $C_{\max}(M)$
- SHAP-permutation Top-K overlap: $A_K(M)$

where:

$$C_{\max}(M) = \max_j p_j(M)$$

- $p_j(M)$ is normalized permutation importance mass.

$$H(M) = - \sum_{j=1}^d p_j(M) \log p_j(M)$$

$$A_K(M) = \frac{|\text{TopK}_{\text{perm}}(M) \cap \text{TopK}_{\text{SHAP}}(M)|}{K}$$

Now define a single deployability score:

$$S(M) = w_R \widetilde{R^2}(M) + w_H \widetilde{H}(M) - w_C \widetilde{C_{\max}}(M) + w_A \widetilde{A_5}(M)$$

Definitions

- $\widetilde{\cdot}$ = normalized to $[0, 1]$ across the 3 models
- $w_R, w_H, w_C, w_A \geq 0$ are weights that sum to 1

This is the missing piece: until weights are declared, “best” is not well-defined.

3) What the data says qualitatively (no hand-waving)

From the artifacts:

- **Epoch 5** has the **highest** surrogate R^2 , but also the **most concentrated** attributions (highest top-feature dominance, lowest entropy). That is the classic “more interpretable but more brittle” pattern for nonstationary trading.
- **Epoch 3** has the **best balance** (highest entropy, lowest top-feature dominance) and strong peer agreement.
- **Epoch 4** is the best on **method-consistency** (SHAP vs permutation Top-5 overlap = 5/5) and has intermediate balance between E3 and E5.

So the choice boils down to:

- If we prioritize **robustness/balance** → **E3**
- If we prioritize **explanation-method invariance** → **E4**

- E5 is the “over-converged” candidate: best surrogate, worst balance → usually not the safest for live short-premium...
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4) A single clear recommendation (the one that matches the report)

Final: Use epoch_3_013026 as the primary deployment model.

Precisely why:

1. It minimizes single-factor dependency by having the **lowest top-feature dominance** and **highest entropy** among the three, which is the most important property for live short-premium robustness (regime shift survivability).
2. It still has strong interpretability (surrogate R^2 is already high enough to extract stable rules).
3. Its feature ranking is highly consistent with the other epochs (high cosine + high rank correlation + high top-K overlap), meaning it is not an “odd one out,” it’s a less brittle version of the same learned logic.

Secondary note (to resolve the “but epoch 4 has perfect SHAP agreement” issue)

Epoch 4’s perfect SHAP-permutation Top-5 agreement is **valuable**, but it is not sufficient to override the robustness criteria unless we explicitly set w_A very high.

So, we use this as a **tie-break / validation gate**, not the main score:

- If Epoch 3 and Epoch 4 are both acceptable on robustness gates, and if you require method-invariance as a compliance constraint, then we can select E4.
 - Otherwise, for pure live survivability of a short-premium system, we select E3.
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5) Make it unambiguous going forward (the rule to bake into the tool)

Set the policy:

Deployment policy (recommended)

1. **Hard gates (must pass):**
 - No leakage (masked features must be zero)
 - No output collapse (variance not near zero for key heads)
2. **Primary score = robustness:**
 - maximize $H(M)$
 - minimize $C_{\max}(M)$
3. **Secondary score = interpretability:**
 - maximize $R^2(M)$
4. **Tie-break = explanation invariance:**
 - maximize $A_5(M)$

This policy yields: **E3 > E4 > E5**, which matches the report's conclusion and removes ambiguity.

Q.E.D.

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